

THE 7 LAYERS OF THE OSI MODEL

Introduction: The **Open Systems Interconnection (OSI) Model** is a conceptual framework developed by the International Organization for Standardization (ISO) to describe how data is transmitted between devices over a network.

The OSI Model consists of **seven layers**, which includes:

1. **Layer 7: Application**
2. **Layer 6: Presentation**
3. **Layer 5: Session**
4. **Layer 4: Transport**
5. **Layer 3: Network**
6. **Layer 2: Data Link**
7. **Layer 1: Physical**

1. LAYER 7: APPLICATION

The **Application Layer** provides network services directly to applications and allows users and applications to interact with network resources.

Functions

- Provides network services to applications.
- Supports web browsing and web applications.
- Supports email communication.
- Provides file-transfer services.
- Allows applications to communicate over networks.

Examples of protocols associated with this layer include HTTP/HTTPS, SMTP and FTP.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
1. SQL Injection (SQL)	An attacker inserts malicious SQL commands into application input to manipulate a database.	Use parameterized queries, prepared statements, input validation and least-privilege database accounts.

2. Cross-Site Scripting (XSS)	An attacker injects malicious scripts into web pages viewed by other users.	Apply output encoding, input validation and an appropriate Content Security Policy (CSP).
3. Cross-Site Request Forgery (CSRF)	An attacker tricks an authenticated user into performing an unwanted action on a website.	Use CSRF tokens, SameSite cookies and appropriate origin/referrer validation.
4. Command Injection	An attacker manipulates application input to execute unauthorized operating-system commands.	Avoid executing shell commands with user input, validate input and use safe APIs.
5. Phishing	Attackers impersonate trusted organizations or individuals to trick users into revealing credentials or sensitive information.	Use security awareness training, MFA, email filtering, URL verification and phishing-resistant authentication where possible.

2. LAYER 6: PRESENTATION

The **Presentation Layer** is responsible for how data is represented, formatted and interpreted between communicating systems.

Functions

- Performs data translation and formatting.
- Handles character encoding and decoding.
- Performs data compression and decompression.
- Supports data encryption and decryption.
- Converts data between different formats.

Protocols used here include SSL/ TLS, X509 and other encoding standards.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. Weak Encryption	Weak or outdated encryption algorithms can allow attackers to recover protected information.	Use modern, approved cryptographic algorithms and regularly update cryptographic configurations.
2. TLS Downgrade Attacks	An attacker attempts to force a connection to use a weaker security protocol or cipher.	Disable obsolete TLS versions and weak cipher suites and enforce modern TLS configurations.
3. Certificate Attacks	Attackers may exploit invalid, fraudulent or improperly validated digital certificates.	Properly validate certificates and maintain trusted certificate authorities and certificate-management processes.
4. Encoding Attacks	Manipulated or unexpected character encoding can cause applications to interpret data incorrectly.	Normalize and validate input and use consistent, secure character encoding.
5. Compression-Based Attacks	Attackers can exploit information revealed through encrypted data compression in certain side-channel attacks.	Avoid compressing sensitive and attacker-controlled data together and use appropriate modern security configurations.

3. LAYER 5: SESSION

The **Session Layer** establishes, manages and terminates communication sessions between applications.

Functions

- Establishes communication sessions.
- Maintains active sessions.
- Terminates sessions when communication is complete.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. Session Hijacking	An attacker obtains a valid session identifier and uses it to impersonate a legitimate user.	Use HTTPS/TLS, secure session tokens and session expiration.
2. Session Fixation	An attacker causes a victim to use a session identifier known to the attacker.	Generate a new session ID after authentication and invalidate old session identifiers.
3. Session Replay	An attacker captures valid session information and reuses it later.	Use short-lived tokens, timestamps, nonces and cryptographic protections.
4. Session Token Theft	Attackers obtain session cookies or tokens through malware, phishing or insecure applications.	Secure cookies with HttpOnly and Secure attributes, use HTTPS and protect endpoints against malware.
5. Session Exhaustion	An attacker creates excessive sessions to consume server resources.	Implement session limits, rate limiting, timeouts and resource monitoring.

4. LAYER 4: TRANSPORT

The **Transport Layer** provides end-to-end communication between applications and controls how data is delivered between devices. Important protocols at this layer include TCP and UDP.

Functions

- Provides end-to-end communication.
- Segments data into smaller units.
- Reassembles data at the destination.
- Provides reliability through TCP.
- Performs flow and congestion control.
- Uses port numbers to identify applications and services.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. SYN Flood	An attacker sends large numbers of TCP connection requests without completing the connections, exhausting server resources.	Use SYN cookies, firewalls, rate limiting and intrusion prevention systems.
2. UDP Flood	An attacker sends large volumes of UDP traffic to consume bandwidth and system resources.	Apply rate limiting, firewall rules and DDoS protection.
3. TCP Session Hijacking	An attacker attempts to take control of an established TCP connection.	Use encryption such as TLS, secure authentication and properly randomized sequence numbers.
4. Port Scanning	An attacker scans a system to identify open ports and potentially vulnerable services.	Use firewalls, close unnecessary ports and monitor network scanning activity.
5. TCP RST Injection	An attacker sends forged TCP reset packets to terminate legitimate connections.	Use encrypted connections, packet filtering and network monitoring.

5. LAYER 3: NETWORK

The **Network Layer** is responsible for logical addressing and routing packets between different networks.

Functions

- Provides logical addressing using IP addresses.
- Determines routes between networks.
- Forwards packets between networks.
- Handles packet routing.
- Supports protocols such as IPv4, IPv6 and ICMP.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. IP Spoofing	An attacker falsifies the source IP address of network packets.	Use ingress/egress filtering, firewalls and anti-spoofing controls.
2. ICMP Flooding	An attacker sends excessive ICMP packets to consume network or system resources.	Configure firewalls and rate-limit unnecessary ICMP traffic.
3. Smurf Attack	An attacker uses spoofed ICMP traffic to cause multiple devices to send responses to a victim.	Disable directed broadcasts and implement anti-spoofing and filtering.
4. Routing Attacks	An attacker manipulates routing information to redirect or disrupt network traffic.	Use secure routing protocols, authentication and route filtering.
5. IP Fragmentation Attacks	Maliciously fragmented packets can be used to bypass security controls or exhaust system resources.	Properly configure firewalls and intrusion-prevention systems and reject malformed fragments.

6. LAYER 2: DATA LINK

The **Data Link Layer** is responsible for communication between devices on the same local network and uses physical addresses such as MAC addresses.

Functions

- Provides node-to-node communication.
- Uses MAC addresses to identify devices.
- Organizes data into frames.
- Detects some transmission errors.
- Controls access to the network medium.
- Supports switching and VLAN operations.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. MAC Spoofing	An attacker changes their device's MAC address to impersonate another device.	Use port security, MAC address controls and network access control.
2. ARP Spoofing/Poisoning	An attacker sends false ARP information to associate their MAC address with another device's IP address.	Use Dynamic ARP Inspection (DAI), DHCP snooping and network segmentation.
3. MAC Flooding	An attacker sends large numbers of fake MAC addresses to overwhelm a switch's MAC address table.	Enable switch port security and MAC address limits.
4. VLAN Hopping	An attacker attempts to gain access to traffic belonging to another VLAN.	Properly configure VLANs, disable unnecessary trunking and use secure native VLAN configurations.
5. STP Attacks	An attacker sends malicious Spanning Tree Protocol information to influence network topology.	Use BPDU Guard, Root Guard and appropriate STP security configurations.

7. LAYER 1: PHYSICAL

The **Physical Layer** is the lowest layer of the OSI Model. It deals with the physical transmission of raw bits over a communication medium.

Functions

- Transmits raw binary data (0s and 1s).
- Defines physical cables, connectors and signals.
- Handles electrical, optical and radio signals.
- Defines data transmission speeds and physical network characteristics.
- Provides the physical connection between network devices.

Common Vulnerabilities/Attacks and Defensive Measures

Vulnerability/Attack	Description	Defensive Measure
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1. Cable Tapping	An attacker physically connects to a network cable to intercept transmitted data.	Use secure cabling locations, physical access controls and encryption.
2. Device Theft	Network equipment such as routers, switches or servers can be physically stolen.	Use locked server rooms, CCTV, access control and equipment tracking.
3. Hardware Tampering	An attacker physically modifies or connects unauthorized hardware to a network device.	Restrict physical access and regularly inspect networking equipment.
4. Jamming	An attacker interferes with wireless radio signals, preventing legitimate communication.	Use appropriate wireless security, frequency management and alternative communication channels.
5. Physical Disruption	Cutting cables, disconnecting equipment or damaging infrastructure can interrupt network connectivity.	Protect cables and network equipment, use redundant connections and implement physical security controls.